

Case 2: Cartesian Path Motion Planning

MoveIt Task Constructor on the xArm 7, E-Waste Battery Extraction, Redback Operations

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1. Purpose

This document records the second motion planning variant. Every stage is solved by the MTC CartesianPath solver, which moves the end effector along a straight line in task space by interpolating poses at a fixed step and solving inverse kinematics at each step [1], [2]. The scene, task stages and execution settings are identical to Case 1.

2. Environment

Component	Version or detail
Host	Windows laptop running WSL2
Operating system	Ubuntu 24.04 LTS (Noble) in WSL2
Middleware	ROS 2 Jazzy Jalisco
Motion planning	MoveIt 2 (Jazzy binary packages)
Task planning	MoveIt Task Constructor (Jazzy binary packages)
Robot package	xarm ros2, jazzy branch, built from source in ~/xarm_ws
Simulation	xarm7_moveit_fake.launch.py with ros2_control fake hardware
Project package	xarm7_mtc_planners, node mtc_planner_compare (C++)

3. Scene and Task Structure

Item	Definition
Battery	Box 70 x 50 x 10 mm at (0.40, 0.00, 0.005) m in link_base, shown in orange
Drop zone	Pad 120 x 120 x 5 mm at (0.30, -0.30, 0.0025) m, shown in green
Planning group	xarm7 with IK frame link_eef, tool pointing down for all pose goals
Execution speed	Velocity and acceleration scaling factor 0.15
Hold between stages	2.0 s pause after every motion and every scene event

#	Stage	MTC stage type	Motion
0	Current state	CurrentState	Reads the live robot state and scene
1	Move above battery	MoveTo (pose)	To (0.40, 0.00, 0.20) m
2	Descend to battery	MoveRelative	0.15 m down in z
3	Vacuum ON	ModifyPlanningScene	Attach battery to link_eef
4	Lift battery	MoveRelative	0.15 m up in z
5	Transfer to drop zone	MoveTo (pose)	To (0.30, -0.30, 0.20) m

#	Stage	MTC stage type	Motion
6	Lower battery	MoveRelative	0.12 m down in z
7	Vacuum OFF	ModifyPlanningScene	Detach battery
8	Retreat	MoveRelative	0.12 m up in z
9	Return home	MoveTo (joints)	All seven joints to 0 rad

4. Planner Configuration

The solver interpolates the tool pose every 10 mm. For pose goals, MoveTo and MoveRelative pass the IK frame to the solver directly. For the joint space goal in Return home, the solver reads the IK frame from its own properties, so it must be set on the solver as well [3].

```

} else if (type == "cartesian") {
  // Case 2: straight line end effector motion
  auto c = std::make_shared<mtc::solvers::CartesianPath>();
  c->setStepSize(0.01);
  geometry_msgs::msg::PoseStamped ik;
  ik.header.frame_id = EEF;
  ik.pose.orientation.w = 1.0;
  c->setProperty("ik_frame", ik);
  s = c;
}

```

5. How to Run

Terminal 1: start the simulated xArm 7 with MoveIt and RViz.

```

source /opt/ros/jazzy/setup.bash
source ~/xarm_ws/install/setup.bash
ros2 launch xarm_moveit_config xarm7_moveit_fake.launch.py

```

Terminal 2: build the package and run this case.

```

cd ~/xarm_ws
MAKEFLAGS="-j1" colcon build --packages-select xarm7_mtc_planners
source install/setup.bash
ros2 run xarm7_mtc_planners mtc_planner_compare --ros-args \
  --params-file ~/xarm_ws/xarm7_params.yaml -p planner:=cartesian

```

Optional parameters: speed (default 0.15), pause (default 2.0 s), countdown (default 5 s) and execute (default true). Every run appends one row to ~/mtc_results.csv.

The parameter file is prepared once, as described in the Case 1 document.

6. Results

Run	Outcome	Plan (ms)	Cost	Exec (s)	Joint path (rad)
1	Failed: missing ik_frame	437.4			
2	Failed: missing ik_frame	406.7			

Run	Outcome	Plan (ms)	Cost	Exec (s)	Joint path (rad)
3	Failed: missing ik_frame	219.7			
4	Success after fix	308.2	11.899	14.647	5.972

Metric definitions: planning time is the wall clock time of `task.plan()`, which includes reading the planning scene from `move_group`. Cost is the MTC solution cost from its default cost function, where lower is better. Execution time is the summed duration of all trajectories, excluding the demonstration pauses. Joint path length is the summed Euclidean distance between consecutive joint space waypoints.

Runs 1 to 3 failed because of configuration, not planning. In each of them, stages 1 to 8 planned successfully as straight line motions and only Return home failed. Setting `ik_frame` on the stage alone did not resolve it; setting it on the solver did.

7. Observations

- **Straight line motion:** the descend, lift, lower and retreat stages keep the tool on a vertical line, which suits approach and retreat above a battery cavity.
- **Determinism:** the solver has no random sampling, so the same start state and goals always produce the same path.
- **Fastest planning in this test:** 308 ms for the full ten stage task, the lowest of the three cases on a single run.
- **Longest motion:** the largest joint path (5.97 rad) and execution time (14.65 s) of the three cases, because holding a straight tool line requires extra joint motion.
- **Limitation:** the solver does not search around obstacles and can fail part way near joint limits or singularities, so it is best reserved for short, clear approach and retreat moves.

8. References

- [1] MoveIt Contributors, "Pick and place with MoveIt Task Constructor," MoveIt 2 Documentation. [Online]. Available: https://moveit.picknik.ai/main/doc/tutorials/pick_and_place_with_moveit_task_constructor/pick_and_place_with_moveit_task_constructor.html. Accessed: Sep. 28, 2026.
- [2] M. Görner, R. Haschke, H. Ritter, and J. Zhang, "MoveIt! Task Constructor for task level motion planning," in Proc. IEEE Int. Conf. Robotics and Automation (ICRA), Montreal, Canada, 2019, pp. 190–196.
- [3] Open Robotics, "moveit_task_constructor API documentation (ROS 2 Jazzy)." [Online]. Available: https://docs.ros.org/en/ros2_packages/jazzy/api/moveit_task_constructor_demo/. Accessed: Sep. 28, 2026.
- [4] MoveIt Contributors, "Move Group C++ Interface," MoveIt 2 Documentation. [Online]. Available: https://moveit.picknik.ai/main/doc/examples/move_group_interface/move_group_interface_tutorial.html. Accessed: Sep. 28, 2026.
- [5] UFACTORY, "xarm_ros2: ROS 2 developer packages for robotic products from UFACTORY," GitHub, jazzy branch. [Online]. Available: https://github.com/xArm-Developer/xarm_ros2. Accessed: Sep. 28, 2026.
- [6] MoveIt Contributors, "MoveIt 2 Documentation." [Online]. Available: <https://moveit.picknik.ai/main/index.html>. Accessed: Sep. 28, 2026.